

FRC Robot

Full-field autonomy for competition robotics.



2025 FRC robot with blue bumpers at the competition



2025 FRC robot and team members in the arena

ROBOTICS

Led development of the 2025 robot. AprilTag-based SolvePnP and wheel encoder data were fused through a Kalman filter for field localization. PID path following tracked trajectories generated with Bezier curves.

2025

Robot development

Sensor fusion / Autonomous navigation / Path planning

[Watch autonomous routine](#)

FIRA Self-Driving Car

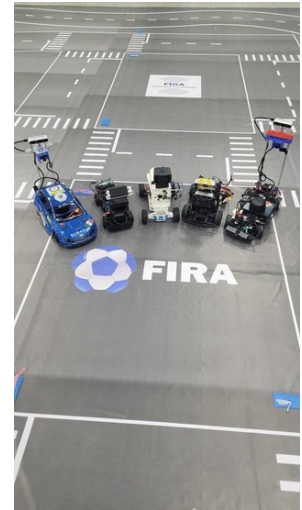
A self-driving car built for the world stage.



FIRA self-driving car on a marked road course



World championship car and robotics equipment



Self-driving cars on the FIRA competition track

AUTONOMOUS VEHICLES

An initial car came together in three weeks. Development continued over two months for the world championship in Brazil, where the car placed second globally.

2nd

Place globally

Self-driving / FIRA RoboCup / World championship

FRC Robot

Vision localization with two cameras.



2024 robot vision localization demonstration



2024 competition robot demonstration

VISION & LOCALIZATION

Developed a localization system using two OV9281 cameras and SolvePnP on AprilTag fiducial markers around the field. Camera mounts were designed in Onshape.

CAD / AprilTag / SolvePnP / Onshape

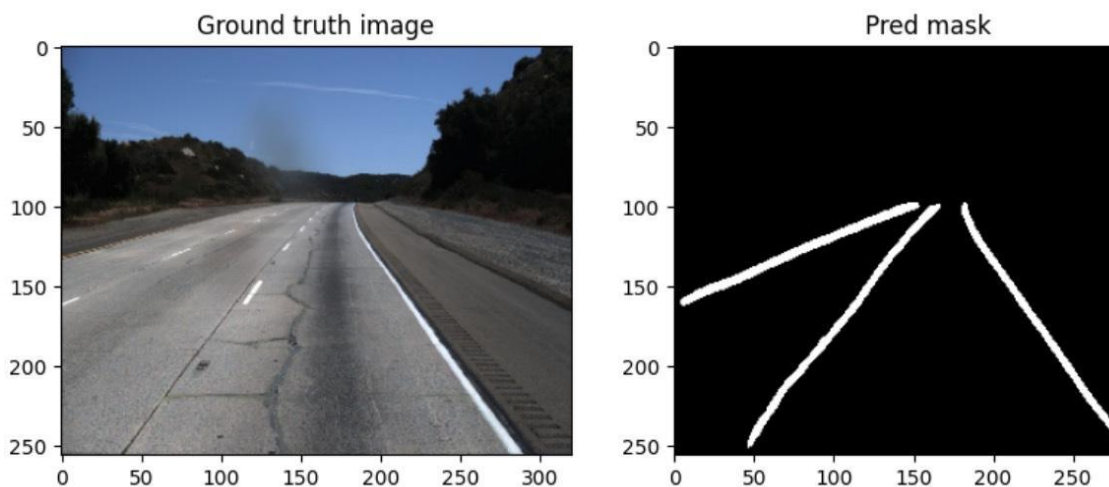
[Watch demonstration 01](#)

[Watch demonstration 02](#)

PROJECT 04 / 10

Mobile CNN Lane Detection

Road perception on a Raspberry Pi 4B.



Road camera image beside the predicted white lane segmentation mask

COMPUTER VISION

A lightweight encoder-decoder network segments lane markings from road camera images. MobileNetV2 provides feature extraction, and a small segmentation head upsamples the features into a lane mask for real-time use.

MobileNetV2 / Semantic segmentation / Raspberry Pi 4B

VEX Robotics

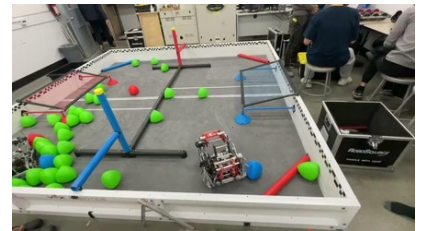
Smooth autonomous navigation from the ground up.



2022 VEX robot on the competition field



2022 VEX robot and mechanism



2022 VEX robot moving on the competition field

ROBOTICS

Designed and programmed a competition robot with wheel encoder localization, PID control, pure pursuit path following, and Bezier curve path generation.

Pure pursuit / Encoder localization / PID control

[Watch demonstration 01](#)

[Watch demonstration 02](#)

March Madness Predictor

Predicting how far a team advances.



MACHINE LEARNING

An ordinal neural network uses engineered efficiency metrics to estimate cumulative round progression probabilities, from the round of 64 to champion. A mean absolute error of 0.72 rounds puts predictions typically within one round of the actual finish; tournament randomness and class imbalance remain limitations.

0.72

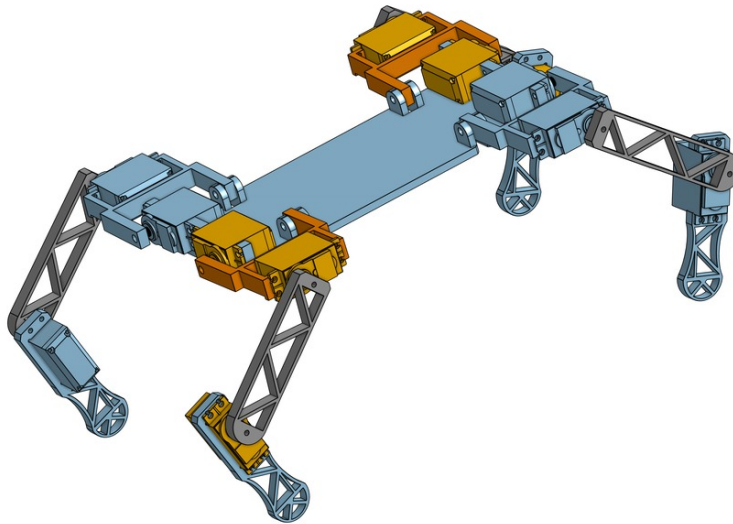
Mean absolute error / rounds

Ordinal neural network / Efficiency metrics / Model evaluation

PROJECT 07 / 10

Robotic Dog

Four legs, one modular chassis.



Onshape CAD model of a four-legged robotic dog with MG996R servo mounts and articulated limbs

MECHANICAL DESIGN

Designed a four-legged robotic dog in Onshape around MG996R servos. Each leg combines a hip joint, femur, and tibia, with dedicated servo mounts connecting the articulated limbs to a central chassis.

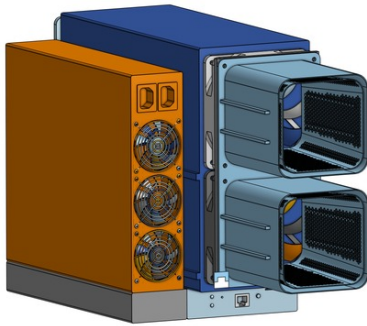
CAD / Onshape / MG996R servos / Quadruped design

[Explore the CAD model](#)

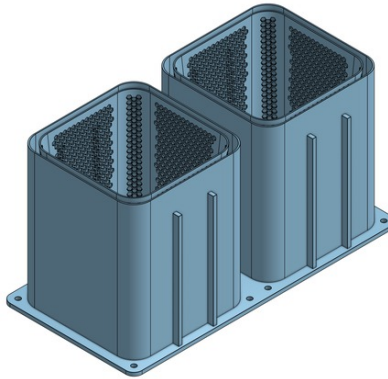
PROJECT 08 / 10

Antminer S19 Pro

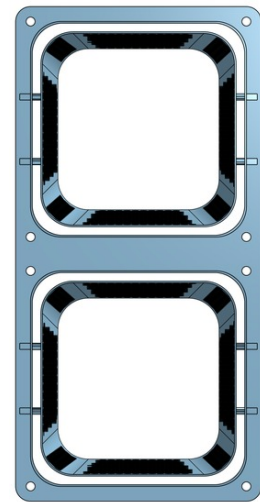
A muffler-inspired approach to fan noise.



Onshape CAD assembly of Antminer S19 Pro with baffled fan ducts



Isometric Onshape view of the isolated fan baffle, showing perforated inner walls, exterior ribs, and mounting flange



View through both fan baffle openings, showing the interior walls and mounting holes

MECHANICAL DESIGN

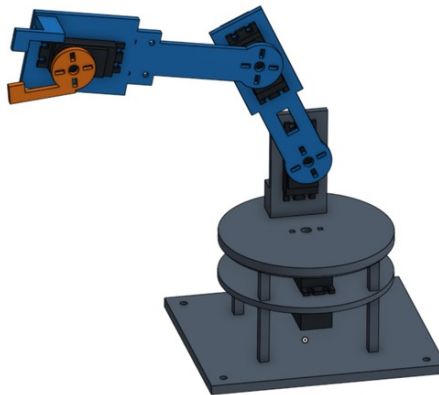
Designed a baffled fan duct for the Antminer S19 Pro in Onshape, inspired by car mufflers. A tapered inner baffle ring is intended to help reduce fan noise.

CAD / Onshape / Baffle design / Fan duct

PROJECT 09 / 10

Robotic Arm

Servo-driven arm design.



CAD model of a robotic arm using MG996R servos

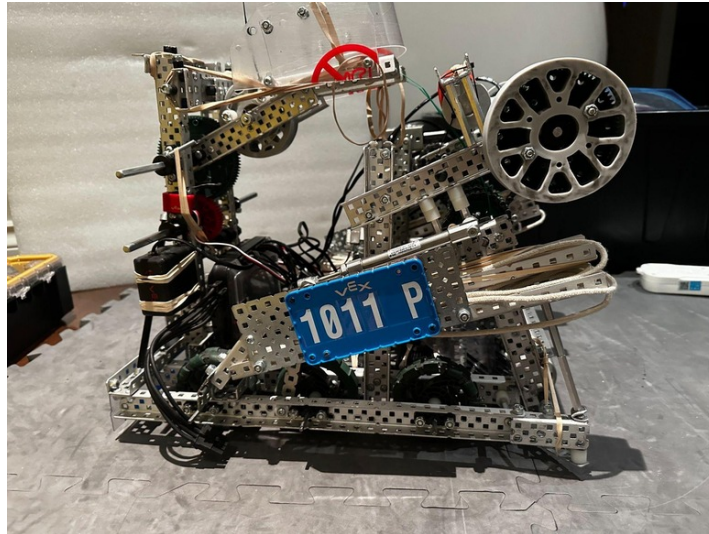
CAD EXERCISE

Designed an articulated robotic arm in Onshape around MG996R servos. The assembly combines a rotating base, linked arm segments, and dedicated servo mounts, with a focus on how the joints and structural parts fit together into a complete mechanism.

CAD / Onshape / MG996R servos / Mechanical design

VEX Robotics

The foundations of autonomous movement.



2021 VEX robot photographed from above

ROBOTICS

Designed and programmed a VEX competition robot using PID control and encoder-based localization for autonomous movement.

Robot design / PID control / Encoder localization